

From Completions to Goals

Building Agents You Can Actually Measure

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Spring AI Community

NYC Java User Group · 2025



Models Got a Lot Better and a Lot Cheaper

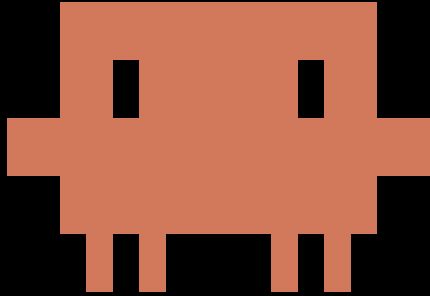
Completions → tool calling → reasoning → planning → agents

"The cost to use a given level of AI falls about **10x every 12 months** ... Moore's law changed the world at 2x every 18 months; this is **unbelievably stronger**."

— Sam Altman, OpenAI CEO · via simonwillison.net

Compute didn't just get faster;
it got **smarter**

Not Moore's Law — Smarty Pants Law



Claude Code

Interactive AI CLI + Unix Composable Utility

- **Interactive mode** — Human-in-the-loop coding with full context
- **Headless mode** — Non-interactive with `-p` flag for automation
- **Unix philosophy** — Pipes, filters, and standard I/O

Chain with other tools:

```
git log --oneline -n 10 | claude -p "Create release notes"
```

The pattern:

```
context | ai | validate
```

From CLI to API

AgentClient Wraps the CLI

- Shift from scripting to code

```
AgentClient client = AgentClient.builder(agentModel).build();

AgentClientResponse response = client
    .goal("Raise coverage to 70%")
    .workingDirectory(workspace)
    .advisors(coverageJudge()) // ← Validates improvement
    .run();

// Agent autonomously: writes tests, runs build, measures coverage
// Judgment: PASS (71.2% ≥ 70%) ✓
```

Same approach as Claude Agent SDK —
wrap the CLI, expose an API

Same Wrapper, Any Agent SDK

AgentClient works with all of them

Claude Agent SDK



Anthropic's Claude Code agent

Gemini Agent SDK



Google's Gemini CLI agent

Amazon Q Agent SDK



AWS's AI development agent

Amp Agent SDK



Amp AI agent platform

Codex Agent SDK



OpenAI's Codex agent

Custom Agent SDK



Bring your own agent

If it runs as a CLI, AgentClient can wrap it

Spring AI Agents

A Complete Agent SDK

Spring AI Community Project · spring.io/blog · rebrand.ly/spring-agents

Not just a CLI wrapper — five components working together:

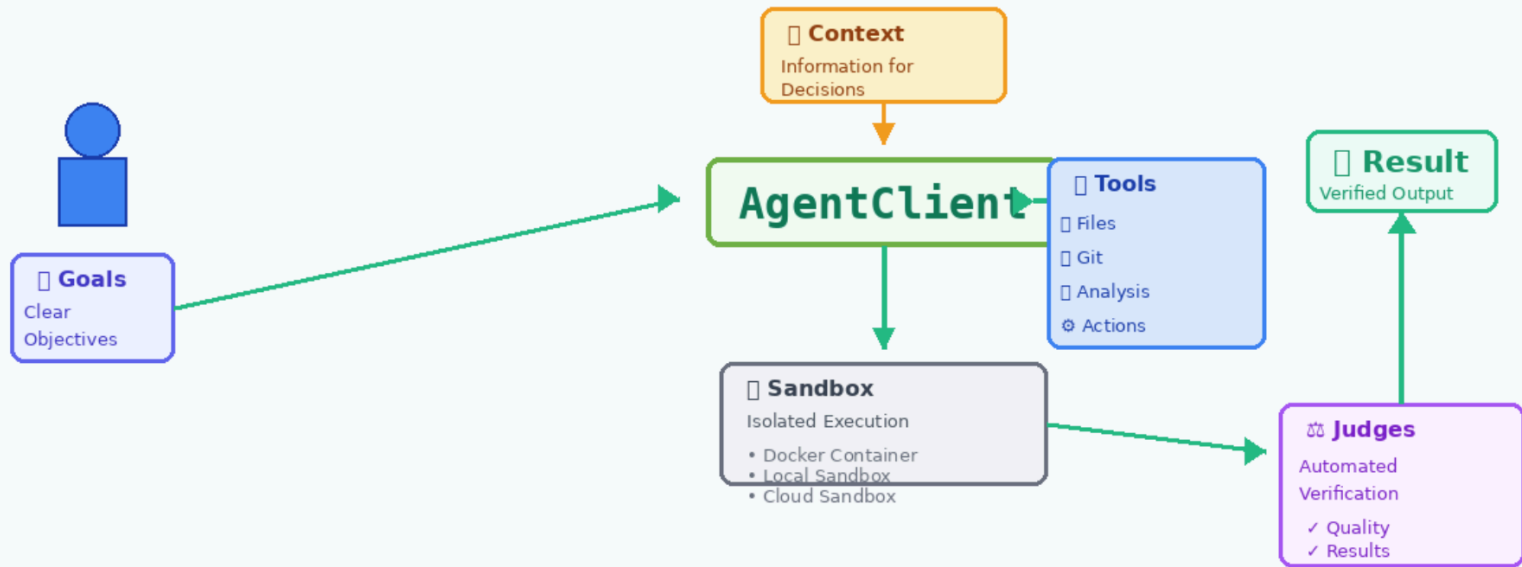
-  **Goals** - Clear objectives agents work toward, optional plans
-  **Tools** - Actions agents can take (CLI, APIs, files)
-  **Context** - Information for decisions (memory, state)
-  **Judges** - Automated verification of completion
-  **Sandboxes** - Isolated execution (Docker or local)



How It All Fits Together

Spring AI Agents: Complete Architecture

Goals • Tools • Context • Judges • Sandboxes



The filesystem IS the pipe

Run Agents on Your Code

Zero Setup Required

- **No cloning, no building** - JBang catalog fetches everything
- **Runs directly on your local codebase** - LocalSandbox by default
- **Switch providers** - Same command, different agent

```
jbang agents@springai coverage \  
  git_url=https://github.com/acme/myapp \  
  target_coverage=70 provider=claude
```

How Do You Know It Works?

Look to the benchmarks

SWE-bench: the de facto standard

"Given a codebase and an issue, generate a patch that resolves the problem"

2,294 real GitHub issues from Python repos · swebench.com

SWE-bench has
problems.

The Reality Gap

SWE-bench Verified:

(SWE-bench original dataset)

Score:

60-75%

SWE-bench Live:

(Fresh dataset)

19%

3x drop

Language Matters:

(SWE-bench Polyglot results)

Python:

~75%

Java:

~7%

10x gap

Score 75% today, 19% tomorrow? That's not a valid test.
And 7% on Java? That doesn't match reality either.

What's Missing

- **Patches only:** No PR review, coverage goals, or compliance checks
- **Python-centric:** 2,294 tasks from 12 Python repos (2023 data)
- **Does not evaluate modern agents:** Claude Code, Gemini CLI, Amazon Q...
- **Very hard to run and reproduce:** Try it yourself!

You can't improve
what you can't measure

Spring AI Bench

Open Benchmarking for Agents

Spring AI Community Project · spring.io/blog · rebrand.ly/spring-ai-bench

- **Multi-agent support**
 - Claude, Gemini, Amazon Q, custom...
- **Enterprise workflows**
 - PR review, coverage, issue labeling, features
- **Reproducible datasets**
 - Standard repos + Docker sandboxes
- **Powered by Spring AI Agents**
 - Uses AgentClient to run and judge agents in sandboxes
- **Extensible to your code**
 - Tune agents on your repos and workflows

If you can't reproduce it, you can't trust it



Real Example

Code Coverage with Two Agents

gs-rest-service · 0% → 71.4% (both agents)

Practice	Claude	Gemini
@WebMvcTest	✓	✗
jsonPath()	✓	✗
BDD naming	✓	✗
Edge cases	✓	✓

Why It Matters:

- **@WebMvcTest** = 10x faster
- **jsonPath()** = cleaner code
- **BDD naming** = readable tests

Same prompt.
Same coverage.
Different quality.

BetterBench

Who Watches the Watchers?

Stanford's quality framework for AI benchmarks · betterbench.stanford.edu

- **The Problem:** Most AI benchmarks fail basic quality standards
 - Not reproducible, contaminated data, opaque methodology
- **The Solution:** 46 criteria across 4 stages
 - Design · Implementation · Documentation · Maintenance
- **Spring AI Bench Alignment:**
 -  Open & transparent - Published code, Apache 2.0
 -  Easy contribution - Community submissions, new tracks
 -  Java-first - Enterprise workflow leaderboards

Think of it as ISO certification for benchmarks

The Paradigm Shift

From Completing Text to Completing Goals

Completions API → ChatClient

Complete sentences, generate text

Agent CLIs → AgentClient

Complete goals, ship features

The tools got smarter.
Now measure if they work.

JetBrains AI Arena



Developer Productivity

AI Arena

An open platform for benchmarking
AI Coding Agents

JetBrains AI Arena



Developer Productivity

AI Arena

The first ever real benchmark for
Java and Spring developers

15+ repositories
140+ realistic coding tasks
Consistent infrastructure



ACP SDK

LSP →
ACP

Open standard.
Plug once,
connect to all.



README

License



Agent/Client Protocol

Agent Client Protocol

The Agent Client Protocol (ACP) standardizes communication between *code editors* (interactive programs for viewing and editing source code) and *coding agents* (programs that use generative AI to autonomously modify code).

Learn more at agentclientprotocol.com.

Integrations

Editors

- [Zed](#)
- Emacs via [agent-shell.el](#)
- [JetBrains](#) (*coming soon*)
- [AionUI](#)
- [marimo notebook](#)

About

A protocol for connecting any editor to any agent

agentclientprotocol.com

- Readme
- View license
- Activity
- Custom properties
- 947 stars
- 6 watching
- 71 forks

[Report repository](#)

Releases 11

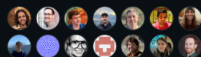
[v0.4.10](#) Latest
3 hours ago

[+ 10 releases](#)

Packages

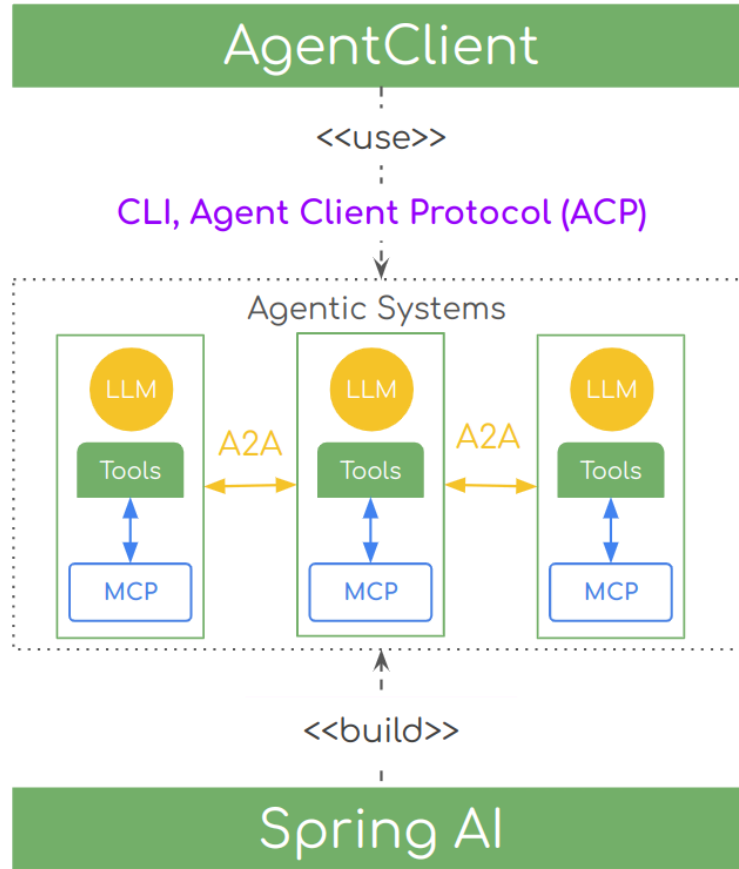
No packages published

Contributors 26



[+ 12 contributors](#)

Languages



Claude Code CLI

Gemini CLI

Embabel

Spring MCP Agent

Spring MemGPT

...

Thank You!

ChatClient → AgentClient



Spring AI Agents

rebrand.ly/spring-agents



Spring AI Bench

rebrand.ly/spring-ai-bench



Community

github.com/spring-ai-community

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